

CoRoT-1b

R-2503

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_210127 · created 2026-08-24T23:34:01+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459241.6613 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.041
Transit depth (Rp/Rs)^2	1.61 +/- 1.04 [%]
Orbital Inclination (inc)	88.93 +/- 2.9
Airmass coefficient 1 (a1)	0.872 +/- 0.04
Airmass coefficient 2 (a2)	0.117 +/- 0.076
Scatter in the residuals of the lightcurve fit is	4.57 %
Best Comparison Star	#1 - [516, 198]
Optimal Method	PSF photometry
Transit Duration (day)	0.0708 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.127 ± 0.041 vs 0.1388 (deviation 0.29σ)
Duration fit vs published	0.0708 d vs 0.1046 d (deviation 5.73σ)
Mid-time O-C	2.9 min
Depth SNR	3.1
Residual scatter	4.57 %

Light curve

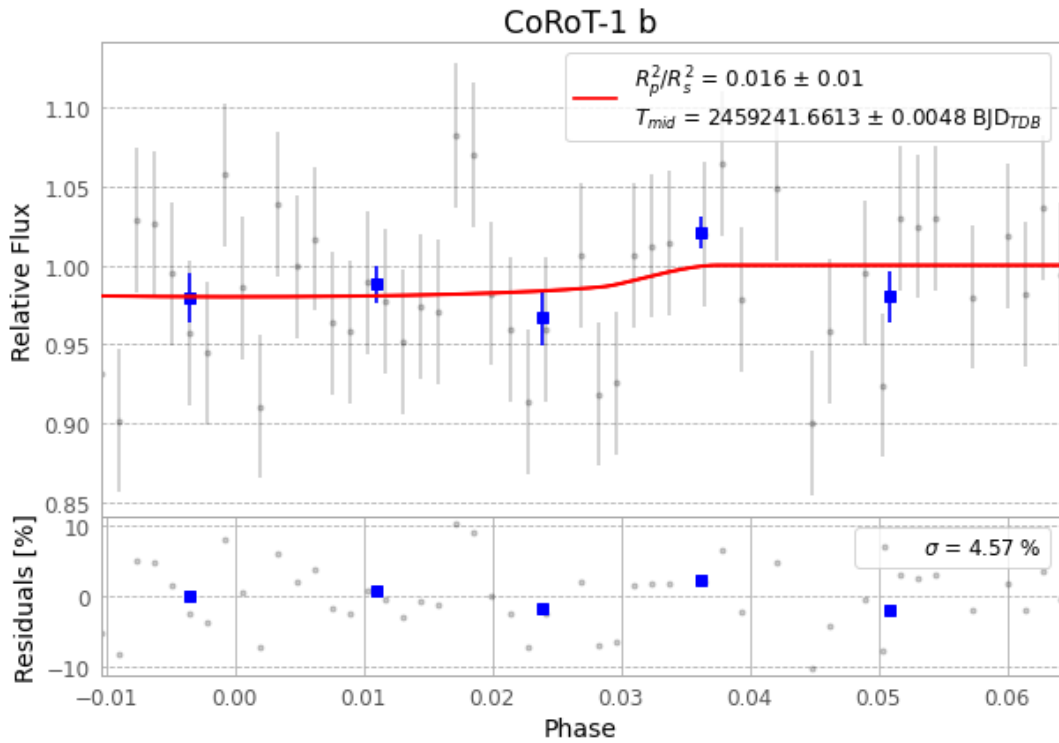


Figure 1 — FinalLightCurve_CoRoT-1 b_2021-01-26.png (SHA-256 0c439fa788b4698e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 283], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f09bba2220a9fb86...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

49 FITS frames (32 MB), CoRoT-1210127032112.FITS → CoRoT-1210127060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-210127.log (6b5bc3f459f2...), FinalLightCurve_CoRoT-1 b_2021-01-26.png (0c439fa788b4...), FinalLightCurve_CoRoT-1 b_2021-01-26.csv (d7e1abd8453e...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)